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Research paper



Engineering material failure analysis report generation based on QWen and Llama2

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ABSTRACT

Engineering material failure carries risks to the public and personal safety and causes economic loss. Failure analysis report describes the causes of failure and gives the improvement strategy, but requiring a lot of manpower of experts. The powerful text understanding and generation capabilities of large language models make automatic analysis possible. This paper presents an automatic report generation method using large language models, including training on only a few samples and instruction fine-tuning method (prompt engineering). The experiments are executed on two popular large language model bases: QWen and Llama2, involving 127 failure reports in 9 fields such as petroleum and petrochemical, nuclear and thermal power, ocean engineering, energy engineering, infrastructure, aerospace, transportation, equipment manufacturing, and water conservancy. Using different questioning ways, GPT evaluation and analysis based on manual verification are performed on the generated results. The experimental results show that the model can generate valuable case reports after being trained with specific domain data. The introduction of artificial intelligence technology for engineering failure prediction can enhance the safety of engineering development, improve industrial production efficiency, reduce labor costs, provide decision support during data analysis, and foster the emergence of new markets and business models.

1. Introduction

1.1. Application of artificial intelligence in engineering material failure analysis

The purpose of engineering material failure analysis is to analyze the cause or critical factors of the failure, take corrective action to eliminate the cause of the failure mode or mechanism, or prevent its recurrence. Engineering material failure analysis typically examines the causes of component failure, the environment in which the component operates, the materials of the failed component, the failure phenomena, and improvement measures. Some researchers focus on a single area, conducting in-depth analysis from one or more of these perspectives. Olajumoke et al. [1]. evaluated the concrete quality in a collapsed building. Zamani et al. [2]. analyzed the metallurgical and mechanical causes of drill pipe failure. Hu et al. [3]. studied the marine corrosion failure mechanism of weathering steel. Xia et al. [4]. and Pang et al. [5]. investigated crack propagation in high-temperature ceramics and functionally graded materials, providing theoretical support for engineering material design. Chou et al. [6]. and Ma et al. [7]. analyzed the causes of wind turbine blade damage and tunnel collapse in ground fissure areas, offering risk management recommendations. Zhang et al. [8]. examined coal pillar instability assessment methods, providing direction for failure analysis in abandoned and deep mines.

With the rapid development of computer technology, applying artificial intelligence methods to failure analysis can greatly improve efficiency and reduce labor costs. Thike et al. [9]. developed an ontology-based hybrid reasoning method, enhancing the effectiveness of failure analysis. Sana et al. [10]. addressed the reuse of waste cooking oil (WCO) and electrode material optimization in electric discharge machining (EDM) by employing artificial neural networks (ANN) and the non-dominated sorting genetic algorithm (NSGA-II) to improve machining performance and environmental sustainability. Jang et al. [11]. proposed a self-learning data-driven method to establish failure criteria. Golmohammadi et al. [12]. analyzed the application of machine learning (ML) models in Materials Science, highlighting successful cases, categorizing relevant literature, and providing guidelines for developing effective ML models to predict material properties. Liu et al. [13].

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proposed a Lang2Sim framework using three types of expert language models to transform textual descriptions into tailored material simulations, addressing the challenge of converting diverse human language inputs into functional and executable simulation code.

With technological advancements, the application of large language models in engineering material failure analysis has become a popular topic. Engineering material failure analysis helps to trace the causes of component failure and avoid risks for engineering services and engineering implementation. Failure analysis relies on rich domain knowledge and requires manual completion by experts, so it is very important to use artificial intelligence to generate failure analysis reports. Previous methods of artificial intelligence were mostly used for fault detection [14], fault classification [15] and fault analysis [16].

Schneller et al. [17]. proposed an optimized neural network method to predict fatigue failure, significantly reducing the need for traditional testing. Zhang et al. [18]. explored the application of convolutional neural networks (CNN) in additive manufacturing to reduce material waste. Gavallas et al. [19]. used CNNs to quickly predict the mechanical properties of composite materials, greatly improving computational efficiency. Sun et al. [20]. proposed a CNN model for fatigue life prediction with enhanced interpretability. Dunn et al. [21]. applied GPT-3 for named entity recognition and relationship extraction in materials chemistry. Sierepeklis et al. [22]. created a thermoelectric materials database, and Polak et al. [23]. introduced a conversational data extraction method, improving automation accuracy and portability in data collection.

1.2. Large language model

Large language model has mainly gone through the early Turing Test, rule-based, statistics-based, and today's deep learning and big data-based language models. Rule-based language models require linguists to analyze language and then process the language into a machine-readable format. This method not only needs to analyze a large amount of language data, reason and predict the language, and establish a mathematical model of natural language context and correlation, but also needs to specify many rules manually to facilitate the understanding of the machine, which not only increases the labor cost, but also cannot avoid the contradiction between the rules. Today, it has evolved into a language model based on deep learning and big data, with a larger corpus and deeper neural networks on which the model can be trained and fine-tuned to achieve the desired functionality.

A Large language model (LLM) is a model trained on large-scale text data with tens of billions of parameters. As can be seen from Fig. 1, the emergence of Transformer completely opens the door to artificial intelligence, and subsequently various language models are deeply affected by the Transformer model structure. Llama-1 [24] is a large language model released by Meta in February 2023, one of the open source models with excellent performance at that time, with four parameter versions: 7B, 13B, 30B and 65B (where "B" stands for billion, which directly reflects the complexity of the model and the depth of understanding of training data). This is its entry into the open-source big model field. Llama 1 is based on the Transformer architecture and is designed to facilitate miniaturization and popularization of large language models (LLMs). Five months later, Meta released the free commercial version Llama-2 in July 2023 [25], with four parameter versions 7B, 13B, 34B and 70B, all of which were open source except for the 34B model. Compared to Llama-1, Llama-2 expands the pre-trained corpus to 2T tokens, doubles the context length of the model from 2048 to 4096, and introduces techniques such as Groupe-Query attention (GQA).

QWen [26] is a natural language processing model for multi-round question and answer, which adopts the method based on knowledge graph and reinforcement learning to realize more intelligent dialogue interaction. With high dialogue understanding and reasoning ability, QWen can generate dialogue according to the dialogue history and context, and the generated answers are not only accurate, but also provide more relevant information and solutions.

LLMs perform well in the fields of natural language processing, dialogue systems, and text summarization, and are widely used in finance [27–29], medical care [30–32], management [33], psychology [34–35], etc. These models exhibit amazing abilities to solve complex tasks, known as "emergent abilities." Studies have shown that by scaling up models, you can significantly improve their performance on downstream tasks, a phenomenon known as "scaling law". At the same time, the development and application of LLM in artificial intelligence is driving the exploration of Artificial General Intelligence (AGI) [36]. Recently, Hu et al. [37]. explored the potential of LLMs in reliability engineering, assessing their ability to address complex engineering problems and their limitations. The study indicated that LLMs show good performance on reliability engineering exam questions with the

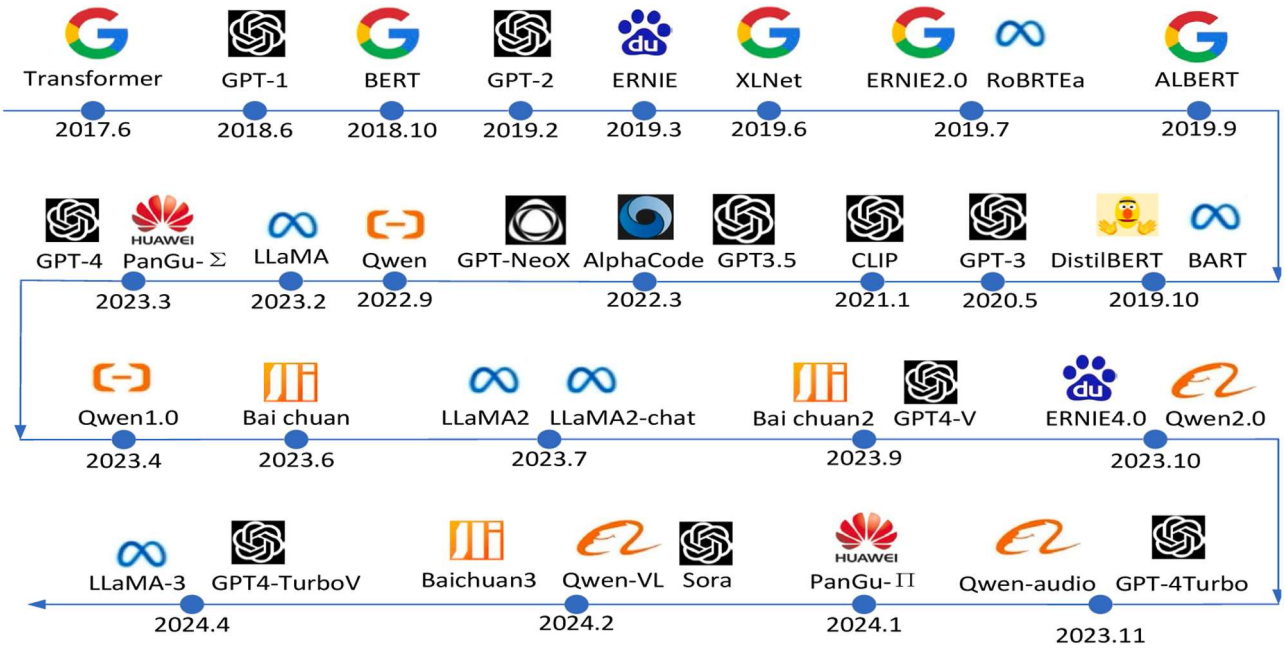


Fig. 1. Technical development path of major language models (2017–2024).

help of prompt engineering. Stewart et al. [38]. conducted the first investigation into the effectiveness of LLMs for Failure Mode Classification (FMC). The study not only demonstrated the effectiveness of prompt engineering on LLMs but also highlighted the importance of high-quality fine-tuning datasets for domain-specific tasks. Jiang et al. [39]. presented a practical guide for using prompt engineering with LLMs to facilitate automatic building energy modeling (ABEM). The results of case studies showed that prompt engineering could effectively generate required building energy models (BEM) using few-shot learning, one-shot learning, and chain-of-thought strategies, with a lightweight LLM performing well in various building applications when provided with an appropriate context window. Qi et al. [40]. used ChatGPT to explore the application of System Theoretic Process Analysis (STPA) in Automatic Emergency Braking (AEB) and Demand Side Management (DSM) systems, investigating the impact of collaboration schemes, input semantic complexity, and prompt engineering on results. The results showed that ChatGPT had reliability issues without human intervention. With careful design, it had the potential to surpass human experts, particularly in STPA-specific prompt engineering. Previous research has shown that although LLMs perform excellently in many fields, their reliability in engineering failure analysis is still questioned, especially in the absence of human intervention. Additionally, while LLMs have the potential to handle complex tasks, they may still have certain limitations in specific domains, such as failure mode classification, particularly the situations lacking high-quality, domain-specific fine-tuning datasets. This paper mainly examines the effectiveness of QWen and Llama2 models in generating engineering material failure reports across nine domains.

The main contributions of this paper are as follows:

(1) To the best of our knowledge, this paper represents the inaugural effort to use pre-trained fine-tuning and prompt engineering based on LLMs to generate failure cases for specific domain tasks.
(2) Use GPT4-based and manual evaluation-based methods to judge the quality of reports generated by LLMs.
(3) Simultaneously, QWen, which supports Chinese, and Llama2, which does not support Chinese, are employed to generate failure cases. Various questioning methodologies are implemented from the perspectives of linguistic differences and prompt engineering techniques, followed by a comparative analysis of the results.

The remainder of this paper is organized as follows: Section 2 provides an overview of the research methodology. Section 3 presents a statistical analysis of the experimental results. Section 4 discusses the issues with the experimental results generated by the QWen and Llama2 models, highlights the limitations of the methods, and addresses the potential impact these limitations may have on the research conclusions. Section 5 offers conclusions and future directions for the research.

2. Methods

As shown in Fig. 2, the case report generation algorithm based on LLM mainly includes data processing, data partitioning, model training and fine-tuning (prompt engineering), report generation and evaluation.

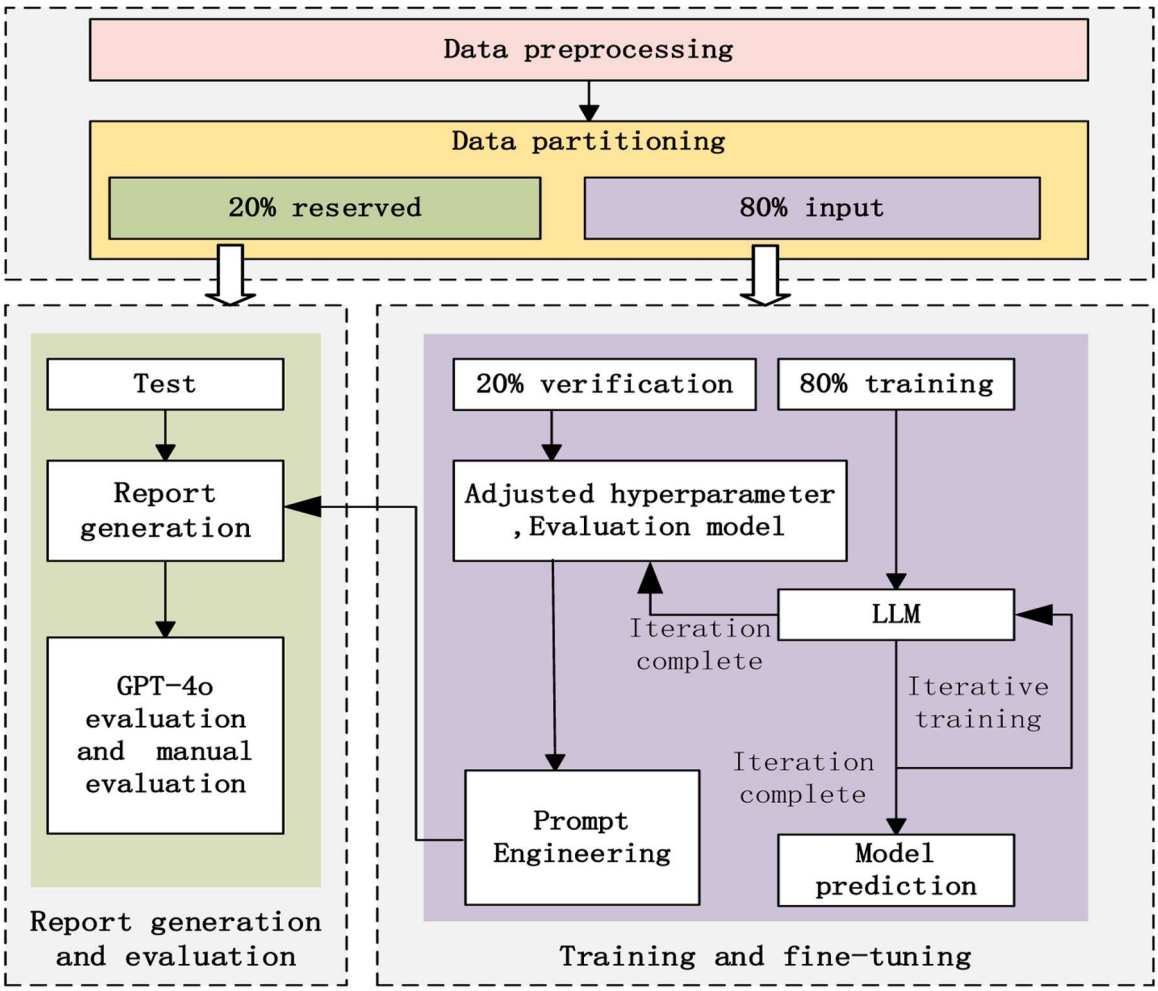


Fig. 2. Flowchart of the method.

2.1. Data processing and partitioning

The original case report is available in both Chinese and English, and the processing is shown in Fig. 3. The data processing process includes data format conversion, data cleaning, special character processing, etc. In the data preprocessing stage, the Chinese and English data to be processed are respectively converted into json file formats that can be recognized by the model.

The purpose of data splitting is to ensure that the model can generalize to new, unseen data and avoid overfitting. This manuscript follows the common data splitting method, which includes a proportional division, where 80 % is used for training, 20 % for testing, and 20 % of the training data is used for validation. The rationale behind this division is as follows:

(1) Avoiding Overfitting: If the model is trained and tested on the same dataset, it may learn noise or specific patterns in the data, leading to poor performance on new data. By splitting the data into training and test sets, we can ensure the model's generalization ability and assess its performance on unseen data.
(2) Model Selection and Tuning: A validation set helps with hyperparameter tuning and model selection. During training, we need to adjust the model's hyperparameters and select the most appropriate model based on the results from the validation set.
(3) Performance Evaluation: The test set is typically used for the final evaluation of the model, ensuring that the evaluation data is different from the training data, thus preventing the test results from being influenced by overfitting during the training process.
(4) Avoiding Data Leakage: Data leakage refers to overlapping or information transfer between the training and test sets, leading to unrealistically optimistic test results. Through data splitting, we ensure the independence of training and evaluation data.

2.2. Training and fine-tuning

In order to realize fine-tuning in specific areas while ensuring the generalization ability of the model, this paper loads the base model and fine-tuning weights through PEFT [41] (Parameter-Efficient fine-tuning), and then combines the two to form a lightweight model for efficient report generation. In addition, this paper also adopts the method of prompt fine-tuning to study the generative ability of the model by changing the prompt words.

2.2.1. Model training

As shown in Fig. 2, 80 % of the data in the input model is further divided, 80 % of the data is used as a training set to achieve model prediction through iterative training, and 20 % of the data is used as a validation set to adjust the model parameters. Implement model evaluation within the model.

2.2.2. Prompt engineering

As shown in Table 1, this paper uses four different prompt words to achieve interactive Q&A with the model. Among them, Q1 and Q2 are Chinese prompt words, and Q3 and Q4 are English prompt words, in order to study the influence of different languages on Qwen model and Llama2 model. The prompt words in Q2 and Q4 are more specific than those in Q1 and Q3, and Q2 and Q4 are based on the keywords in the real report as prompts, so as to explore the impact of the complete expression of prompt words in the prompt project on the model. In addition, this paper takes the case title text as the key content in the prompt word to distinguish the test data in different fields. The titles of twenty-three cases are shown in Table 2.

Prompting refers to the context or instructions provided to a model in natural language processing tasks to guide the model in generating the desired output. In LLMs (such as GPT-3, BERT, etc.), the choice of prompt is crucial because it determines how the model understands the task and directly impacts the quality and relevance of the output.

In many generative or reasoning tasks, selecting an appropriate prompt can clarify the task's objective. We use a prompt such as 'Please generate a case analysis report on ******' to provide information about the task's background and context, which includes keywords related to the engineering failure event itself, as well as the goal of generating a

Table 1
Different questioning ways.

Question ID	The way of question
1	请生成一个"******"案例分析报告;
2	请生成一个"******"案例分析报告, 包括"(1)案例名称(2)失效过程描述(3)服役环境(4)失效构件与材质(5)构件失效形貌(6)失效原因分析(7)造成的影响和危害(8)结论和改进措施"八项内容;
3	Please generate a case analysis report of "******"
4	Please generate a "******" case analysis report, including "(1) case name (2) failure process description (3) service environment (4) failure component and material (5) failure appearance (6) failure cause analysis (7) Impact and harm (8) Conclusion and improvement measures" eight items.

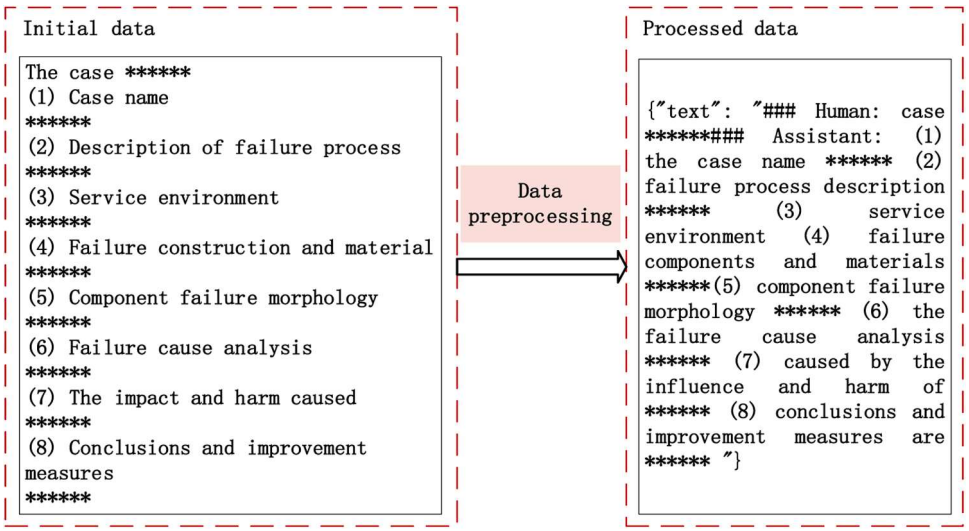


Fig. 3. Data processing example.

Table 2
Case name for the generated report.

ID	Case Report	Domain
1	Plate cracking of crude methanol detachable plate condenser	petroleum and petrochemical
2	Plate heat exchanger plate cracking	
3	Failure of BR heat exchanger in a factory	nuclear and thermal power
4	ammonia corrosion on steam side of copper tube of condenser in Liaocheng Power Plant	
5	copper pipe corrosion of condenser in Harbin Third Power Plant	
6	Bohai Sea anti-ice jacket platform failure	ocean engineering
7	offshore platform lifting point failure	
8	jar spline mandrel fracture	energy engineering
9	Double-shoulder drill pipe joint cracking	
10	longitudinal cracking of Kelly pipe protection joint	
10	reinforced concrete corrosion of Haiyan Bridge pier column	infrastructure
12	Subgrade subsidence of Southern Shaanxi Highway	
13	high cycle fatigue fracture of LPT4 stator blade of an engine	aerospace
14	fatigue fracture of missile steering gear bus line	
15	hydrogen embrittleness fracture of spring gasket for a missile position indicator	
16	nose landing gear fracture of a certain type of aircraft	
17	Damage of concrete pavement of Hening Road	transportation
18	Highway asphalt pavement cracking, depression and mud pumping	
19	Road damage from Changping to Yanqing Section of National Highway	
20	Several cases of accidents caused by insufficient mechanical strength of imported transformers	equipment manufacturing
21	Loader rear axle reducer drive gear shaft tooth failure	
22	Gouhou Reservoir dam failure	water conservancy
23	Palpanxia Dam sulfate erosion	

case analysis report. The secondary prompt further refines the task guidance based on human needs, helping the model better understand these needs. This can improve model performance, guide the generation process, reduce irrelevant content, and enhance the precision and relevance of the output. For specific tasks like failure case analysis, outlining the expected content can increase the model's flexibility, guide the model to generate diverse answers, or reduce bias in the generation.

Among them, the case name is the requirement question, as shown in Table 2, and the other parts are the generation question.

2.3. Report generation and evaluation

2.3.1. Report generation

As shown in Fig. 4, this paper inputs questions into four models according to four ways of asking questions, corresponding to four types of generation cases respectively.

2.3.2. Evaluation

Methods based on GPT-4o and manual verification are adopted in this paper to evaluate the report generation capability of the model. The evaluation process based on GPT-4o is shown in Fig. 5.

Finally, 20 % of the retained data will be used to generate case reports based on LLMs. As shown in Table 3, GPT-4o and the manual verification method have been evaluated across multiple aspects [42], such as language logic, data accuracy, and professionalism in industry practices.

3. Experimental results and analysis

In 3.1, this paper introduces the statistics and division of data in nine fields, as well as the experimental results in the appendix. In 3.2, the results of all original reports and generated reports are evaluated from five perspectives based on GPT-4o. In 3.3, the case of "Plate cracking of crude methanol detachable plate condenser" is evaluated in detail by using questions of "Question 2" and "Question 4" based on manual calibration.

3.1. Data statistics and results

The failure causes of the components are affected by the materials, production and construction, design and environmental factors. The main forms of failure are shown in Table 4. The data in this paper cover nine domains, in which training set and verification set account for 80 % of the total data of subclassification, and test set accounts for 20 % of the total data of subclassification.

The original analysis reports, the generation analysis reports of the Qwen model as the base model, the generation analysis reports of the fine-tuning Qwen model as the base model, the generation analysis reports of Llama2 model as the base model, and the generation analysis reports of the fine-tuning Llama2 as the base model are included in Appendix 1, Appendix 2, Appendix 3, Appendix 4, and Appendix 5, respectively, and GPT-4o-based evaluations are summarized in Appendix 6, as shown in Table 5. Full results are available at https://github.com/jessi-chang/Data_Engineering-failure-analysis.

3.2. Case analysis of generation original model and fine-tuned model

This section focuses on visualizing the original analysis report and the analysis report generated by different models based on GPT-4o evaluation results.

As mentioned in 2.2.2, Q1 and Q2 are Chinese prompt words, while Q3 and Q4 are English prompt words. Meanwhile, Q2 and Q4 are more specific than Q1 and Q3 prompt words. As can be seen from Fig. 6, when QWen model and Llama2 model perform the report generation task in Chinese and English, Qwen model will give valuable information under

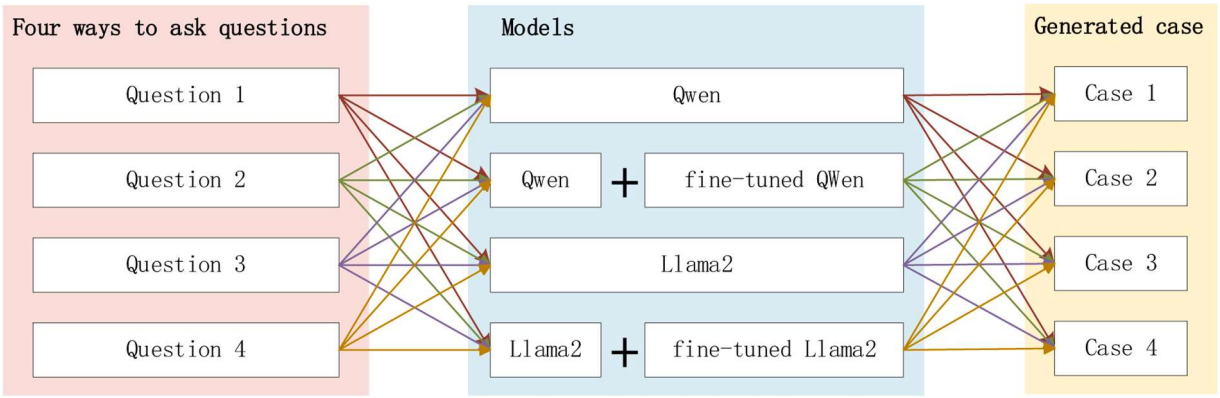


Fig. 4. Case report generation process.

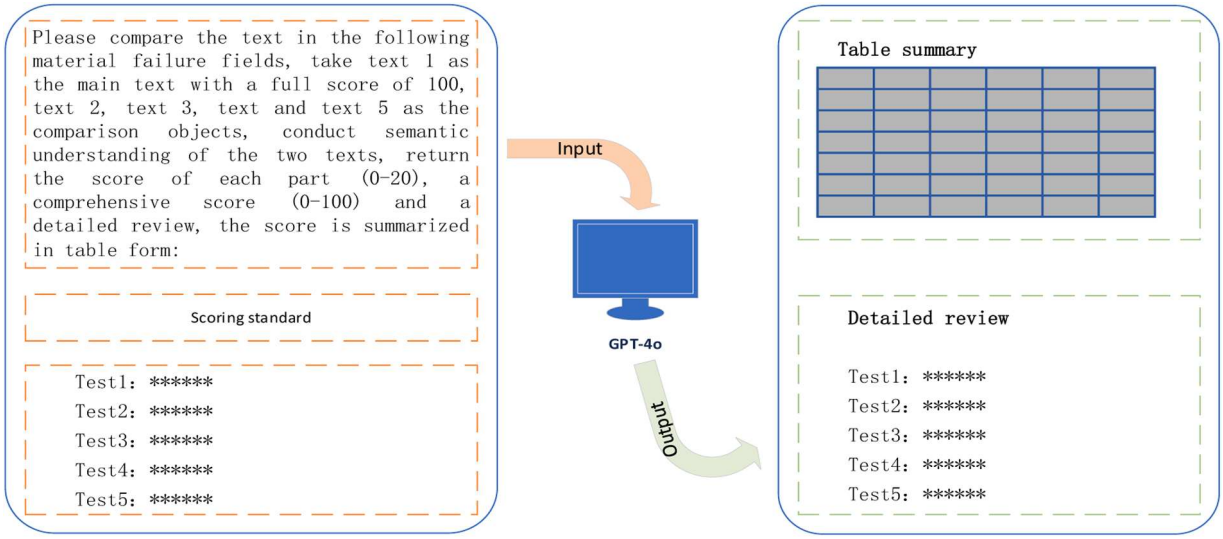


Fig. 5. Scoring process based on GPT-4o.

Table 3
Evaluation criteria.

Number	Evaluation criteria	Explain in detail
1	Depth and breadth of content	Evaluate the coverage and depth of each text in terms of material failure. Including but not limited to: failure types, mechanisms, influencing factors, solutions, etc.
2	Accuracy and reliability	Evaluate whether the content of each text is based on scientific and engineering facts, whether the sources and data cited are credible, and whether there is sufficient experimental or theoretical support.
3	Logical and organized	Evaluate each text for clarity of structure, logical rigor, and reasonable paragraph transitions and conclusions.
4	Technical terms and expressions	Evaluate whether the terminology used in each text is accurate, appropriate for the target audience's level of understanding, and clear.
5	Innovation and uniqueness	Evaluate whether each text presents new ideas, methods, or research results in the field of material failure, and whether it has unique insights.

the prompt engineering in both languages, while Llama2 model will only give valuable information under the prompt engineering in English, which is mainly influenced by the base model itself. This paper is only in petroleum and petrochemical, nuclear and thermal power, ocean engineering, energy engineering, Perform prompt projects in both languages simultaneously on Llama2 model.

According to the boxplot in Fig. 6, it is found that when Q1 and Q3 are used to question the model, the lower extremum of Ft-Qwen model is smaller, which means that the model learns the features it does not want to learn. In addition, the paper also includes overfitting due to the small amount of fine-tuned data. And lack of data diversity, among other reasons, leads to poor model performance.

Based on the median of the boxplot, it is found that when Chinese (Q1, Q2) is used as the prompt word, the rating given by GPT-4o is not significantly lower in Ft-Qwen model than in Qwen model. When English (Q3, Q4) is used as the cue word, GPT-4o gives a significantly higher rating in Ft-QWen model than in Qwen model, but a significantly higher rating in Ft-Llama2 than in Llama2 model.

Based on the upper extremum of the boxplot, it is found that in both Qwen model and Llama2 model, the fine-tuned model performs better than the non-fine-tuned model on the whole, especially in the specific questions (Q2, Q4).

Table 4
Data statistics and data division in nine fields.

Industry field	Major failure mode	Case Statistics (127)			
		Train	Val	Test	Total
Petroleum petrochemical	Corrosion, fatigue, fracture and creep	8	2	2	12
Nuclear power, thermal power	Corrosion, leakage, fatigue, breakage and wear	17	4	3	24
Ocean engineering	Uniform corrosion, pitting corrosion, crevice corrosion, cavitation corrosion, fracture, protective coating failure and ice vibration failure	5	1	2	8
Energy engineering	Corrosion and overload fracture	12	3	3	18
Infrastructure	Subgrade subsidence, pavement damage, rail damage and concrete failure	5	1	2	8
Aerospace	Fatigue, fracture and corrosion	14	3	4	21
Transportation	Fatigue, wear, corrosion, and aging	11	3	3	17
Equipment manufacturing	Early failure, brittle failure, fatigue failure, wear failure, seal failure and corrosion	6	1	2	9
Water conservancy project	Dam cracks, leakage, wear, cavitation, corrosion and aging	6	2	2	10

Table 5
Models and experimental results.

Model	Results
original analysis reports	Appendix 1
QWen-7b	Appendix 2
QWen-7b (Fine-tuning)	Appendix 3
LLama2–7b	Appendix 4
LLama2–7b (Fine-tuning)	Appendix 5
GPT-4o evaluation summary	Appendix 6

All appendices are available at https://github.com/jessi-chang/Data_Engineering-failure-analysis.

Based on the distribution of data in the boxplot and the valid anomaly scores, it can be seen that the quality of report generated by Qwen model is relatively stable regardless of whether it is fine-tuned or

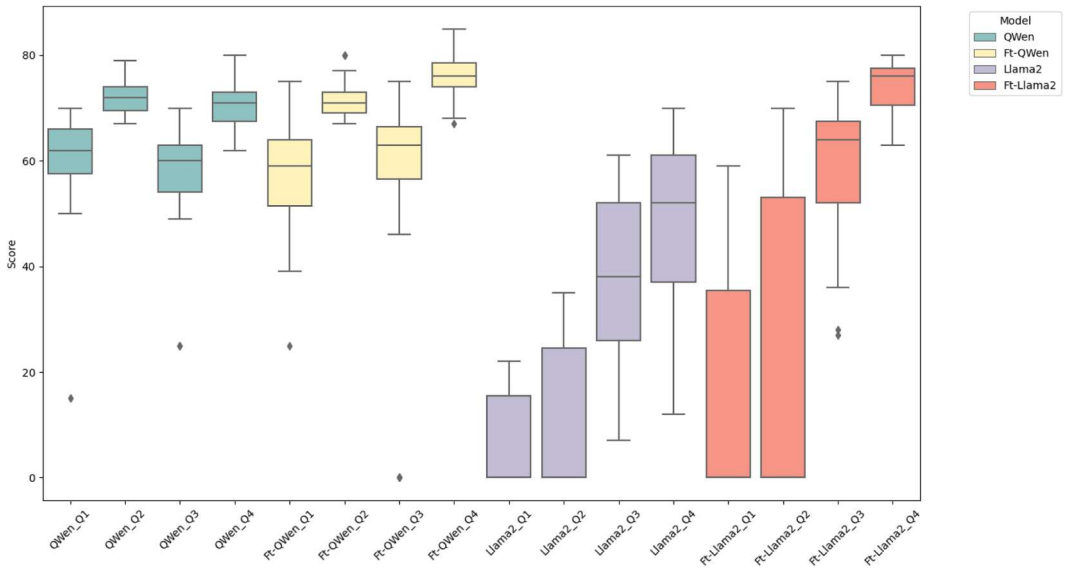


Fig. 6. The performance of the case reports under four types of questioning and two models.

not. However, after fine-tuning, the score distribution of generated reports has been improved overall, especially in Ft-Llama2 model.

Comparing Q1 and Q2 in Chinese and Q3 and Q4 in English, it is found that more precise and detailed questions returned better results in both Chinese and English. It should be noted that since Qwen model and Llama2 model have formulated a series of rules to protect human beings, when the model is asked to generate engineering material failure cases, it may be judged by the model as not conforming to the specified behavior. The required report cannot be output. At this situation, GPT-4o will evaluate this situation as 0 points.

As shown in Fig. 7, we visualize the scores for 23 cases based on four different types of questions. By comparing Q1, Q2, Q3, and Q4, we find that GPT-4o's scores for the reports do not differ much between QWen and the fine-tuned QWen. In contrast, there is a significant difference in GPT-4o's scores between Llama2 and the fine-tuned Llama2, with the fine-tuned Llama2 showing a significant improvement in report generation ability. When comparing Q1 and Q2 in Chinese and Q3 and Q4 in English, we find that more precise and detailed questions lead to better outcomes in both Chinese and English.

3.3. Comparison between the generated case and the original case of the original model and the fine-tuning model

In view of the analysis in 3.1 and 3.2, it can be concluded that specific Question forms have a positive effect on failure cases, and considering the influence of both Chinese and English on the generation results, this section mainly analyzes the Question ways based on "Question 2" and "Question 4", compares the generated cases between different models, and observed the model performance.

3.3.1. Comparison of QWen generation cases with fine-tuned QWen

As shown in Table 6, by comparing the real case of "Plate cracking case of crude methanol detachable plate condenser" (1.1 in Appendix 1), the generation case based on QWen (1.1.1, 1.1.2, 1.1.3, 1.1.4 in Appendix 2) and the generation case based on fine-tuned QWen (1.1.1, 1.1.2, 1.1.3, 1.1.4 in Appendix 3), a summary of the key points under the two ways of asking questions, it can be concluded that:

(1) When the question is asked in the way of "Question 2", the cases generated by both models will divide the generated text into multiple items according to prompt words and introduce data features in the answer. In the original case, the temperature at the hot side is 65 °C, the temperature at the outlet is <38 °C, the pressure is 0.25 ~ 0.3 MPa, and the working medium is 99 % methanol. The working temperature at the cold side is 28 ~ 34 °C, the working temperature at the outlet is 32 ~ 36 °C, the pressure is 0.1Mpa, and the working medium is circulating water. The answer of Qwen model indicates that the temperature is above 30 °C and the pressure is about 10MPa. The answer of the fine-tuned Qwen indicates that the temperature is from −10 °C to +40 °C, the pressure is from 0.1 to 0.2 MPa, and the medium is crude methanol. Thus, the answer of the fine-tuned Qwen is closer to the original case.

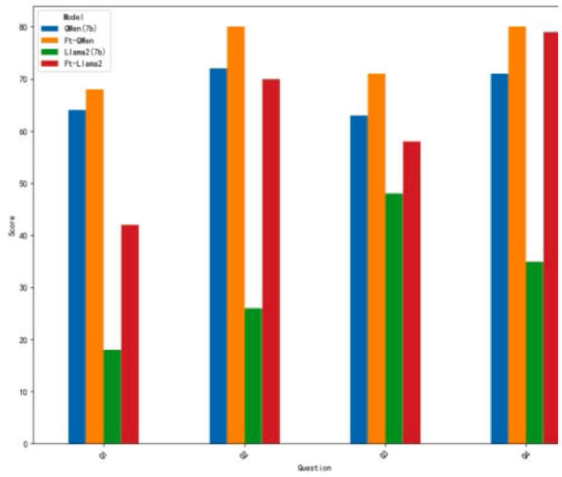
(2) When the question is asked in the way of "Question 4", the cases generated by both models will divide the generated text into multiple items according to prompt words and introduce data features in the answer. However, the answer given by Qwen model for the environment of the failed parts is that the working temperature should be maintained at 80 °C and the pressure at 10 bar, while the answer given by the fine-tuned Qwen for the environment of the failed parts is that the maximum working temperature is 180 °C and the maximum working pressure is 10mpa, which is far different from the original case.

It can be seen from different questioning methods that more accurate questioning will lead the model to return more accurate results. However, no matter which questioning method is used, the fine-tuned QWen trained with professional data will produce more reasonable results than the non-fine-tuned QWen.

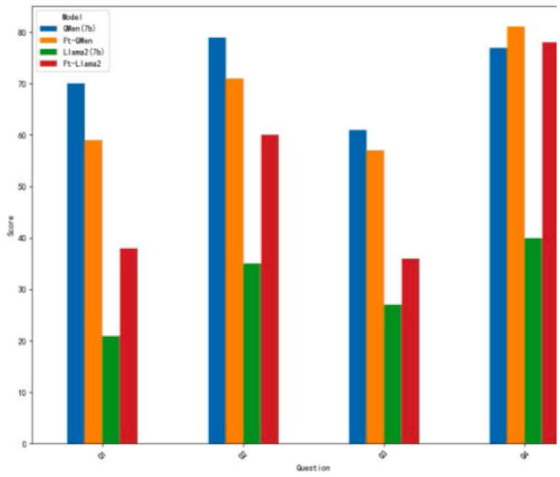
3.3.2. Comparison between the generation cases of QWen and llama2

As shown in Tables 6 and 7, by comparing the real case of "Plate cracking case of crude methanol removable plate condenser" (1.1 in Appendix 1), the generation case based on QWen (1.1.1, 1.1.2, 1.1.3, 1.1.4 in Appendix 2), and the generation case based on Llama2 (1.1.1, 1.1.2, 1.1.3, 1.1.4 in Appendix 4). Summary of the key points under the four two of asking questions, it can be concluded that:

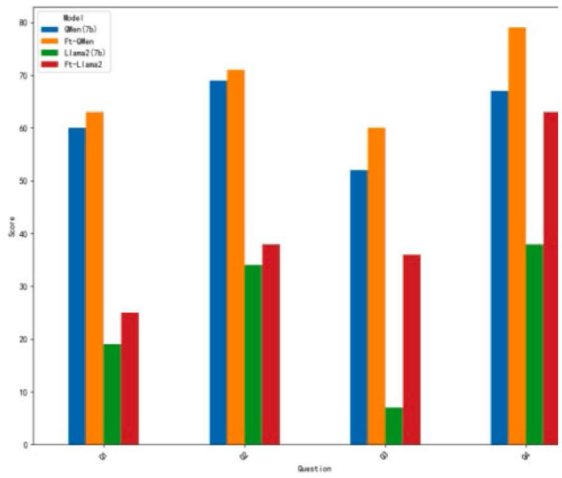
(1) When the question is asked in the way of "Question 2", the generation case of QWen model will divide the generated text into several items according to the requirement of prompt words, and expand the narration one by one, while adding data information. However, the text generated by Llama2 model cannot fully show the reliable content, so this paper argues that the adaptability of Llama2 model to language is weak.



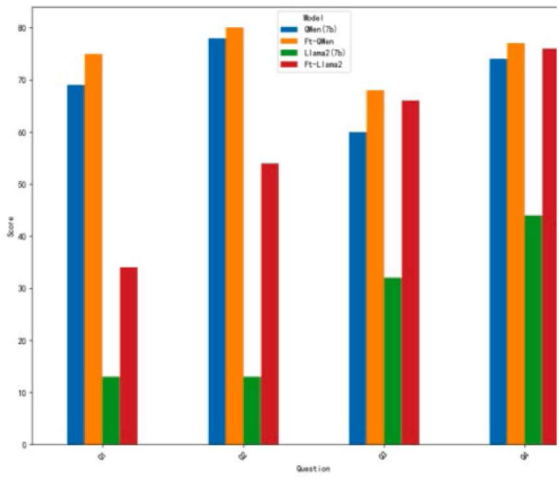
(a) The performance of the case report_1



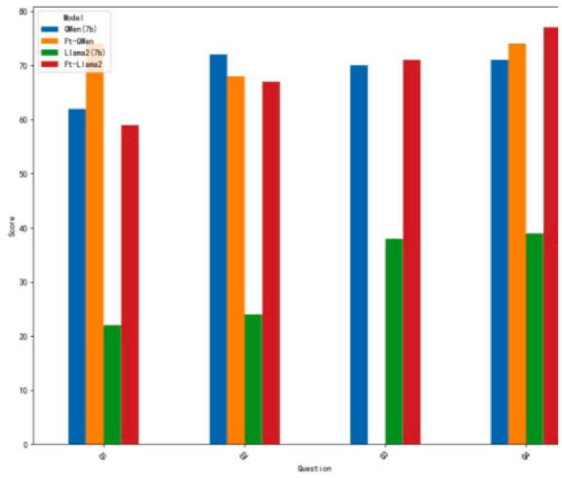
(b) The performance of the case report_2



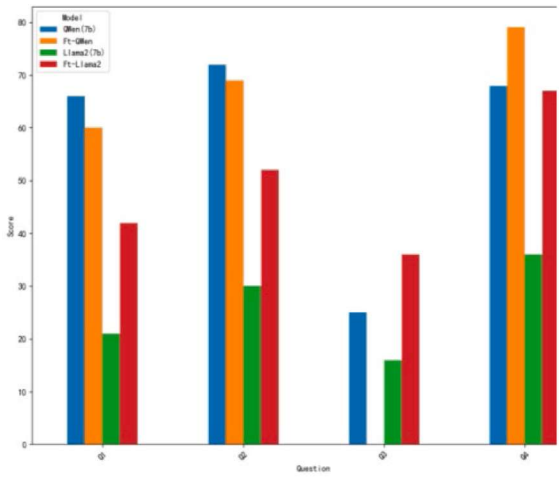
(c) The performance of the case report_3



(d) The performance of the case report_4

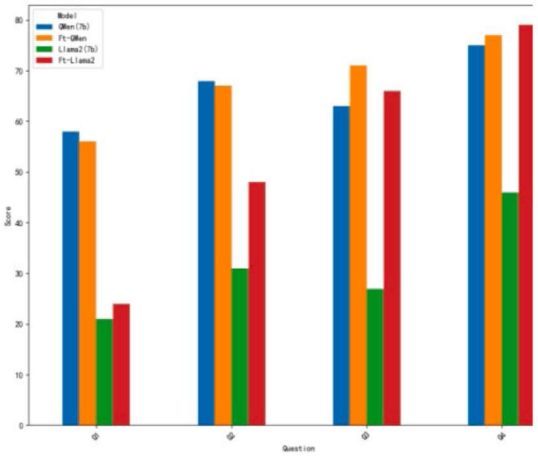


(e) The performance of the case report_5

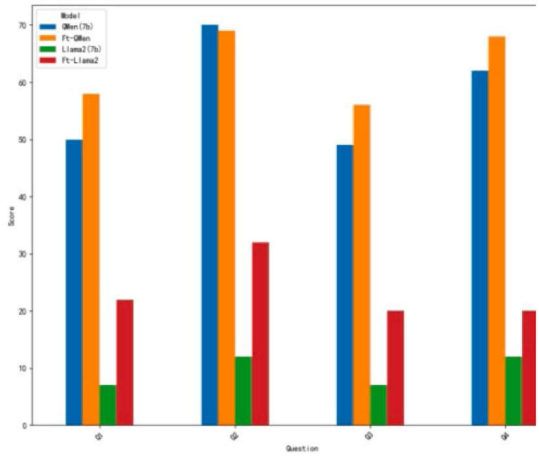


(f) The performance of the case report_6

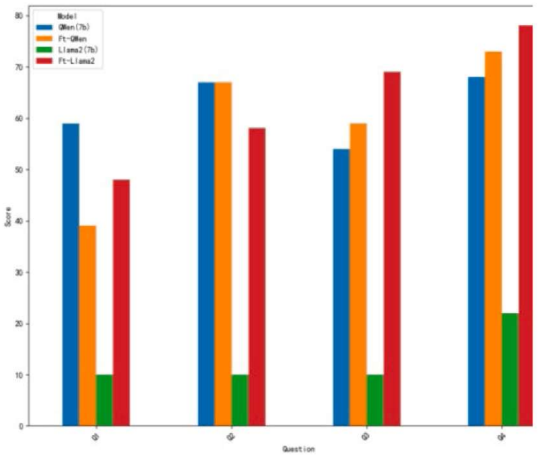
Fig. 7. The performance of the case reports under four types of questioning and different models
(a) and (b): petroleum and petrochemical, (c)-(e): nuclear and thermal power, (f) and (g): ocean engineering, (h)-(j): energy engineering, (k) and (l): infrastructure, (m)-(p): aerospace, (q)-(s): transportation, (t) and (u): equipment manufacturing, (v) and (w): water conservancy.



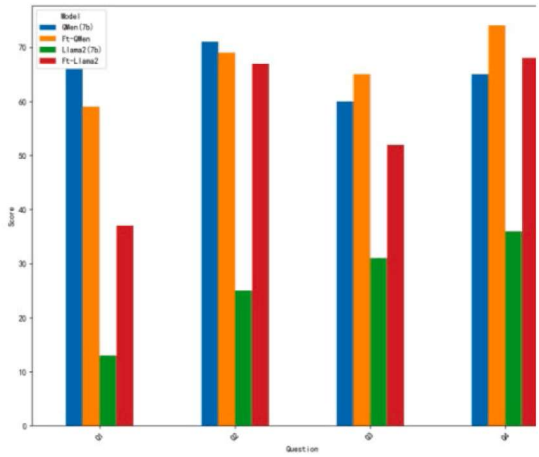
(g) The performance of the case report_7



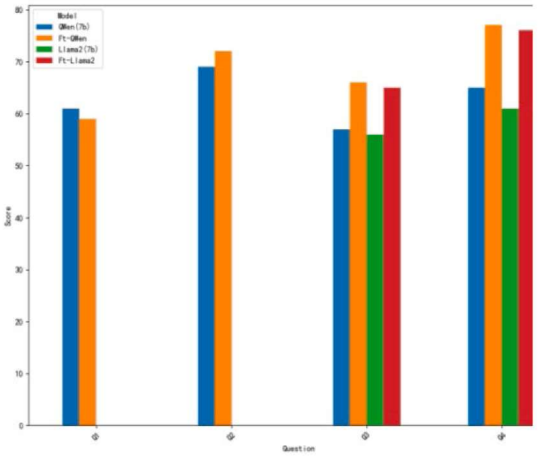
(h) The performance of the case report_8



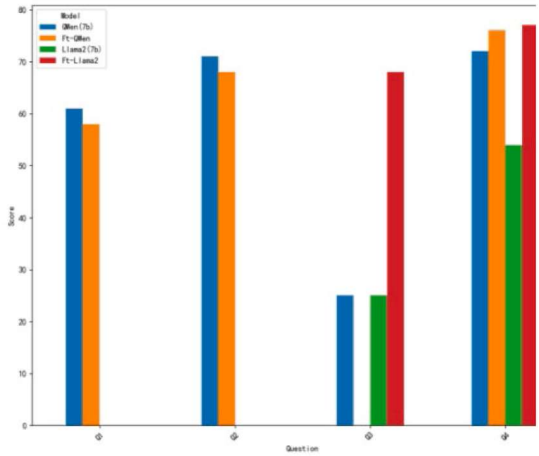
(i) The performance of the case report_9



(j) The performance of the case report_10



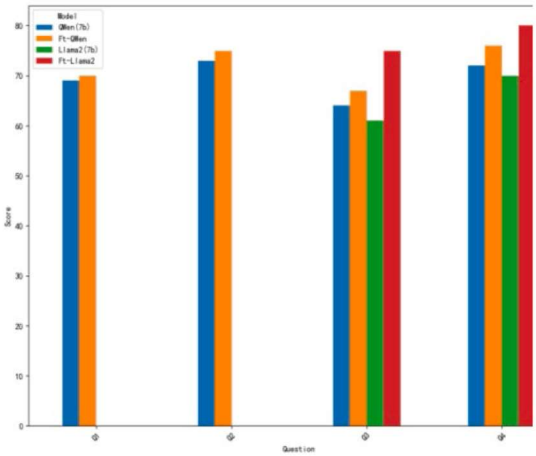
(k) The performance of the case report_11



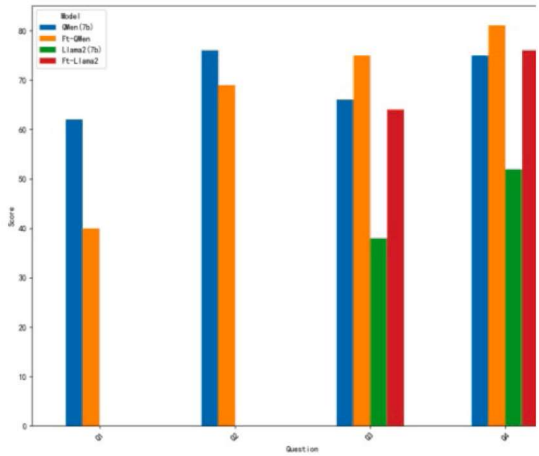
(l) The performance of the case report_12

Fig. 7. (*continued*).

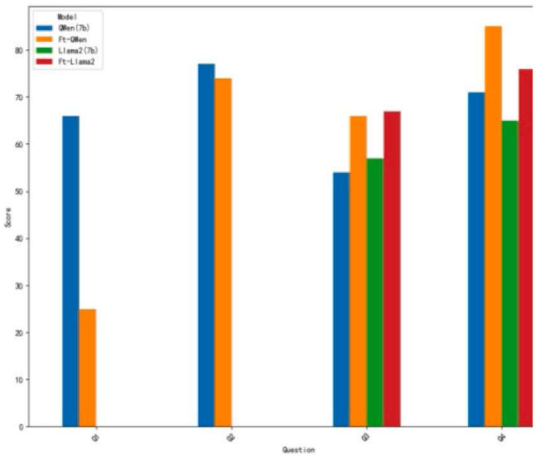
(2) When asking questions in the way of "Question 4", this paper finds that the generation case of QWen model would divide the generated text into several items according to the requirements of the question, and carry out a generalized description of each content, but there was still a gap with the content of the original case. The



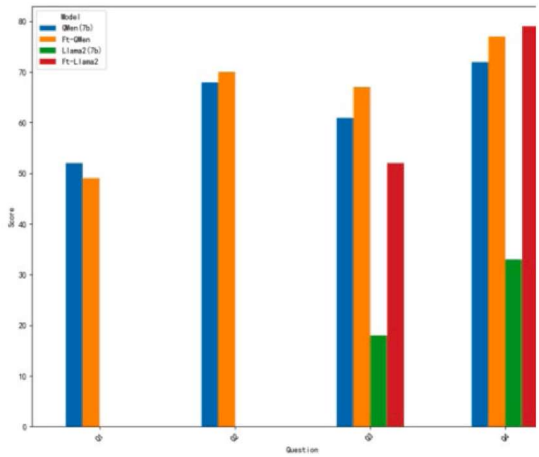
(m)The performance of the case report_13



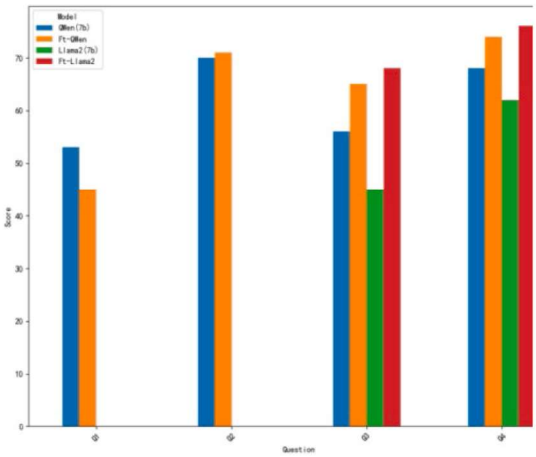
(n) The performance of the case report_14



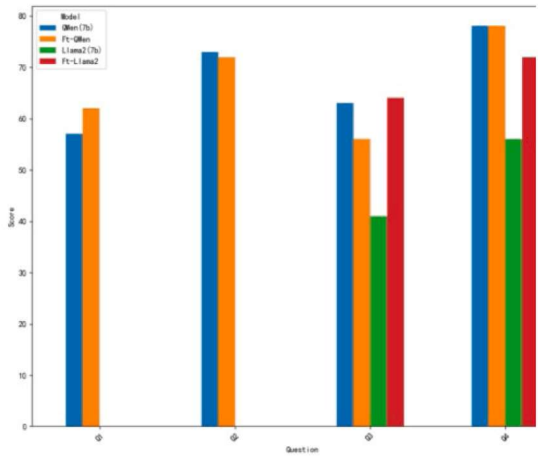
(o) The performance of the case report_15



(p) The performance of the case report_16



(q) The performance of the case report_17

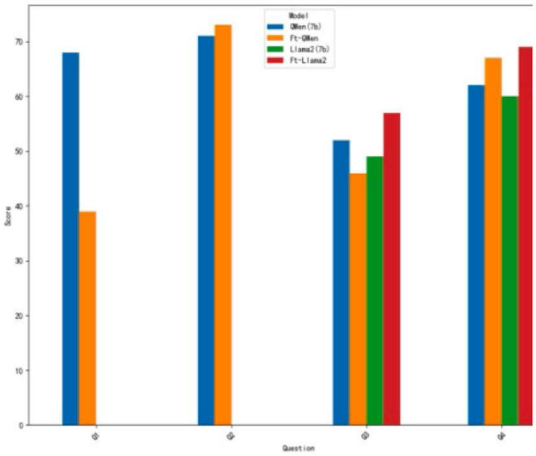


(r) The performance of the case report_18

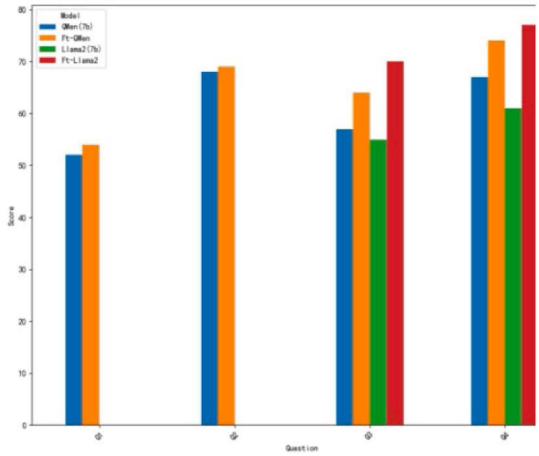
Fig. 7. (*continued*).

generation case of Llama2 model obviously yields more content, but it is still not fully displayed.

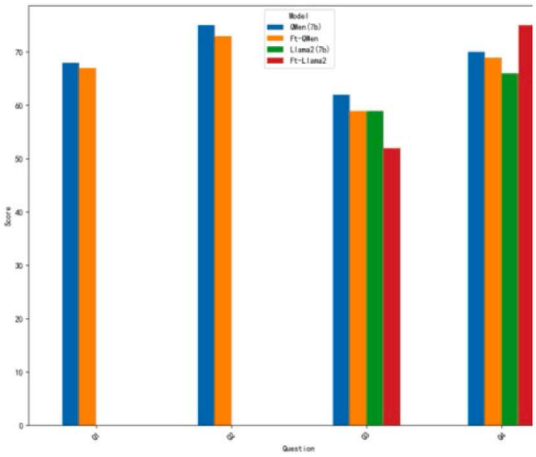
It can be seen from the different questioning methods that Qwen model can generate more reasonable forms of content regardless of the language in which it is asked, while Llama2 model is more sensitive to



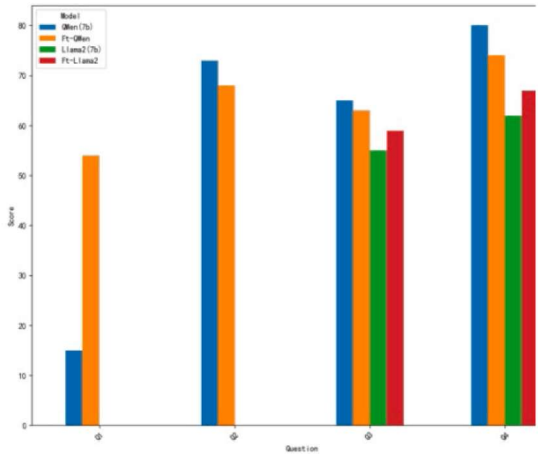
(s) The performance of the case report_19



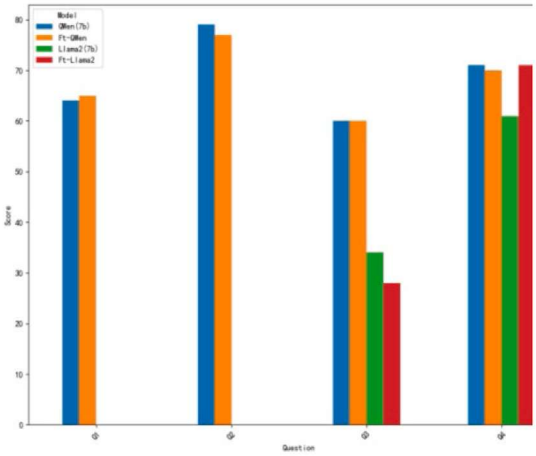
(t) The performance of the case report_20



(u) The performance of the case report_21



(v) The performance of the case report_22



(w) The performance of the case report_23

Fig. 7. (*continued*).

language. While asking questions in English can produce better content, the reports generated by this approach fall far short of expectations. Neither of the two untrained base models can meet our quality requirements for automatic report generation.

Table 6
The comparison of the cases generated by QWen and Ft-QWen in Q2 and Q4 with real cases.

Case	Real case	QWen(Q2)	QWen(Q4)	Ft-QWen(Q2)	Ft-QWen(Q4)
Name of case	Plate Cracking of Crude Methanol Removable Plate Condenser	粗甲醇可拆板式冷凝器板片开裂	Plate Cracking of Crude Methanol Removable Plate Condenser	粗甲醇可拆板式冷凝器板片开裂	Plate Cracking of Crude Methanol Removable Plate Condenser
Form of failure	Crack	开裂	crack	开裂	crack
Service environment	Temperature: hot side inlet 65 °C, outlet <38 °C; The cold side inlet is 28 ~ 34 °C and the exit is 32 ~ 36 °C Pressure: hot side atmospheric pressure; The cold side is 0.25 to 0.3 MPa Working medium: 99 % methanol and mixture on the hot side; Cold side circulating water	温度:高温(30 °C以上) 压力:高压(10Mpa) 工作介质:无	Temperature: High temperature (80 °C) Pressure: High pressure (10 bar) Working medium: None	温度:−10 °C至+40 °C; 压力:0.10.2MPa; 工作介质:粗甲醇	Temperature: 180 °C; Pressure: 10 mpa; Working medium: Crude methanol
Materials	Material: TA1 Component specification: Heat exchange area 100m^2	材质:不锈钢 部件规格:0.8mm	Material:Stainless steel Component specifications: Approximately 1 millimeter thick with a diameter of 1 meter	材质:不锈钢S316L 部件规格:长1.2m,宽0.9m,厚0.06m	Material:stainless steel grade 304 L; Component specifications:Plate thickness 10 mm, size 1000×1000mm
Failure morphology	The fracture analysis of the methanol condenser plate reveals corrosion pits, grain detachment, and intergranular crack propagation. Cracks initiate from corrosion pits, accompanied by needle-shaped corrosion products accumulating at the pit's bottom, with secondary micro-cracks forming during propagation. This indicates structural damage and the underlying corrosion mechanism in the material.	板片表面可见明显的裂纹, 裂纹从边缘向中心延伸, 有的甚至贯穿整个板片.	Mainly located near the edges and corners. These cracks are deep and wide, some even extending to the entire plate. Cracks are accompanied by surface roughness and pitting, indicating corrosion.	板片表面有明显的腐蚀坑蚀, 部分板片上可见明显的裂纹, 裂纹长度一般为几十到几百毫米, 深度可达几毫米至十几毫米.	The cracks were observed to be shallow and narrow, with lengths ranging from several millimeters to several centimeters. They were located near the corners of the plates and extended into the plate thickness.
Cause of failure	Stress corrosion cracking: the metallographic structure of the component is martensitic α' phase +α phase, which increases the brittleness, organizational stress and cracking sensitivity of the material	材质疲劳,设计缺陷,使用不当	Thermal Stress; Mechanical Load; Corrosion.	(1)板片设计不合理; (2)材料选择不当; (3)工艺条件控制不严	Cracks may be stress corrosion cracking (SCC), occurring under high temperature and low dissolved oxygen concentration conditions. In addition, the presence of impurities in crude methanol, such as sulfur-containing compounds, may accelerate the propagation rate of cracks.
Improvement measures	Do not choose titanium or titanium alloy	针对之前分析, 提出解决措施	(1) Consider the effects of thermal expansion and mechanical loads, as well as the possibility of corrosion (2) Select high-quality materials	针对之前分析, 提出解决措施	(1) Regularly inspect and clean the condenser. (2) Using materials with higher SCC resistance.

3.3.3. *Comparison of llama2 generation cases with fine-tuned llama2*

As shown in Table 7, by comparing the real case of "Plate cracking case of crude methanol detachable plate condenser" (1.1 in Appendix 1), Llama2-based generation case (1.1.1, 1.1.2, 1.1.3, 1.1.4 in Appendix 4), and the generation case based on fine-tuned Llama2-based generation case (1.1.1, 1.1.2, 1.1.3, 1.1.4 in Appendix 5). A summary of the key points under the two ways of asking questions, the conclusion can be drawn:

(1) When asked in the form of "Question 2", Llama2 model cannot give a complete answer. Formally, the responses of the fine-tuned Llama2 are presented in English, which further indicates that Llama2 model is more constrained by language. In terms of content, the answer of the fine-tuned Llama2 indicates that the cracks are formed by micro-fatigue cracks and stress corrosion cracks, which is similar to the analysis of failure causes in the original case.
(2) When the question is asked in the way of "question 4", Llama2 model tries its best to give a complete report according to the request of the question, but it still cannot generate a reasonable report. The fine-tuned Llama2 produces a dramatic improvement over the previously generated low-quality text. First of all, from a formal point of view, the generated content can be accurately divided into multiple items according to the requirements of the question. Secondly, from the point of view of the content, the same case name and cracking failure form as the original case are generated, and it is pointed out that the failure process has the influence of alloy materials, which is reflected in the improvement measures in the original case. From the data information of the generated text, the generated data of methanol flow, temperature and pressure are similar to the original case, and there is a white crystal state in the failure form.

It can be seen from different questioning methods that the quality of text generated by Llama2 model is not high, especially when facing Chinese questioning, the content is even worse. After fine-tuning Llama2 model, the model performance has been greatly improved, but it is still limited by the base language, so in this aspect of the test, when asking more detailed questions in English, Llama2 model shows good content.

3.3.4. *Comparison of the generated cases of the fine-tuned QWen and the fine-tuned llama2*

As shown in Tables 6 and 7, by comparing the real case of "Plate cracking case of crude methanol removable plate condenser" (1.1 in Appendix 1), the generation case based on fine-tuning QWen (1.1.1, 1.1.2, 1.1.3, 1.1.4 in Appendix 3), and the generation case based on the fine-tuning Llama2 (1.1.1, 1.1.2, 1.1.3, 1.1.4 in Appendix 5). Summary of the key points under the two ways of asking questions, the conclusion

Table 7
The comparison of the cases generated by Llama2 and Ft-Llama2 in Q2 and Q4 with real cases.

Case	Real case	Llama2 (Q2)	Llama2 (Q4)	Ft-Llama2 (Q2)	Ft-Llama2 (Q4)
Name of case	Plate Cracking of Crude Methanol Removable Plate Condenser	粗甲醇可拆板式冷凝器板片开裂	Plate Cracking of Crude Methanol Removable Plate Condenser	Plate Cracking of Crude Methanol Removable Plate Condenser	Plate Cracking of Crude Methanol Removable Plate Condenser
Form of failure	Crack	无	crack	Crack	crack
Service environment	Temperature: hot side inlet 65 °C, outlet <38 °C; The cold side inlet is 28 ~ 34 °C and the exit is 32 ~ 36 °C Pressure: hot side atmospheric pressure; The cold side is 0.25 to 0.3 MPa Working medium: 99 % methanol and mixture on the hot side; Cold side circulating water	温度:无 压力:无 工作介质:无	Temperature: None; Pressure: None; Working medium: None	Temperature: Gradually increases due to insufficient cooling and pressure buildup. Pressure: Increases due to the expansion of refrigerant (Freon) as it evaporates. Working medium: Coolant, specifically Freon.	Temperature: 50–60 °C (condenser outlet temperature control range) Pressure: Pressure reduction ≤ 0.25 MPa Working medium: Crude methanol with a flow rate of ≥ 600 m^3/h
Materials	Material: TA1 Component specification: Heat exchange area 100m^2	材质:无 部件规格:无	Material:None; Component specifications:None	Material: Sheet metal. Component specifications: Manufactured using TIG welding technique.	Material: Alloy (condenser plate), seamless steel pipe (condenser tube) Component specifications: Condenser plate: FX-80S-60, thickness: 0.8 mm. Condenser tube: ASME SB467–1998, standard: ASTM A335–02, wall thickness: 1.6 mm
Failure morphology	The fracture analysis of the methanol condenser plate reveals corrosion pits, grain detachment, and intergranular crack propagation. Cracks initiate from corrosion pits, accompanied by needle-shaped corrosion products accumulating at the pit's bottom, with secondary micro-cracks forming during propagation. This indicates structural damage and the underlying corrosion mechanism in the material.	无	None	The failure presents as cracks on the chiller panel, which appear due to micro fatigue cracking and stress corrosion cracking, primarily along the weld joints.	The failure appears as a white crystallization state on the surface of the cracks on the condenser plates, indicating brittle fracture.
Cause of failure	Stress corrosion cracking: the metallographic structure of the component is martensitic α' phase +α phase, which increases the brittleness, organizational stress and cracking sensitivity of the material	无	The crude methanol removal plate of crude methanol removable plate condenser is cracked.	Increased pressure within the chiller due to refrigerant expansion. Micro fatigue and stress corrosion cracking at the weld joints. Poorly controlled refrigerant levels, leading to inefficient cooling and pressure buildup. Potential defects in the welding process, which may introduce micro-defects and contribute to crack propagation.	Defective alloy materials lack sufficient toughness, leading to brittle failure and susceptibility to cracking. Non-standard material specifications lack adequate strength and toughness. Insufficient hardness during welding, with local overheating causing hardness to fall below the specified HB600–650, leading to cracking.
Improvement measures	Do not choose titanium or titanium alloy	无	None	Using materials that can withstand high pressure and load. Regularly testing and regulating the coolant and pressure level to maintain system safety.	Use higher-quality alloy materials, such as Ni-Cr-Mo alloy.

can be drawn:

(1) When the question is asked in the form of "Question 2", the fine-tuned QWen will return the required seven items according to the question demand, and indicate the size and stainless steel material of the failed part; The temperature of the service environment is −10 °C to +40 °C, the pressure is 0.1 ~ 0.2 MPa, and the medium is crude methanol. Analyze the cause of the failure and return to the solution. The fine-tuned Llama2 answers Chinese questions in English. From a formal point of view, the model transformation work has been done inside the model. From a content point of view, the generated text is numerous, redundant, and lacks logic.
(2) When asked in the form of "Question 4", the paper concludes that both models generate content under the current model and show the best results under the current test. From a formal point of view, both models return the required seven items in the form of questions, and each item is described in detail. In terms of content, both models generate the case name of the original case. In the failure process description, the original case has the description of time characteristics and cracking phenomenon, which are explained in the generated text of the fine-tuned Llama2, while the generated text of the fine-tuned QWen only describes the cracking phenomenon. In the service environment of the original case, the inlet temperature of the hot side of the service environment is 65 °C, the outlet temperature is <38 °C, the pressure is 0.25 ~ 0.3 MPa, and the working medium is 99 % methanol, etc. The inlet working temperature of the cold side is 28 ~ 34 °C, the outlet working temperature is 32 ~ 36 °C, the pressure is 0.1Mpa, and the working medium is circulating water. The generated text of the fine-tuned QWen points out that under high temperature and high pressure, the maximum working temperature is 180 °C and the maximum working pressure is 10mpa; the generated text of the fine-tuned Llama2 points out that the methanol flow rate is ≥ 600m3/h, the pressure reduction is ≤ 0.25 MPa, and the temperature control range of the condenser outlet is 50–60 °C. The generated text of the fine-tuned Llama2 is closer to the original case. Along the failed components and materials, the material of the original case is TA1 and the component specification is 100m2 heat exchange area. The material in the generated text of the fine-tuned

QWen is 304 L and the specification is 1000 × 1000mm. In the generated text of the fine-tuned Llama2, the condenser sheet is alloy FX-80S-60 with a thickness of 0 .8mm. The condenser tube is made of alloy with a wall thickness of 1 .6mm. According to the above phenomenon, it can be concluded that the fine-tuned Llama2generates richer text. Among the failure morphology of the components, corrosion pits, grain shedding and cracks were found in the original case. The morphology and location of cracks were described in the generated text of the fine-tuned QWen, while the crystalline state of the crack surface and the failure type of brittle fracture were described in the generated text of the fine-tuned Llama2. Compared with the original case, the generated text of the two models has some advantages.

As can be seen from the different questioning ways, both the fine-tuned QWen and the fine-tuned Llama2 have improved over their respective base models, but Llama2 model is more sensitive to language. The fine-tuned QWen shows superior performance when asked in the "Question 2″. Both fine-tuned models have merit when asked in the "Question 4" format, but the fine-tuned Llama2 comes out on top when analyzed point by point.

4. Discussion

Since Llama2 model is very sensitive to language, when questions are asked using "Question 1" and "Question 2", garbled data will be displayed. Therefore, Appendices 4 and 5 will no longer provide results for results for the five areas of infrastructure, aerospace, transportation, equipment manufacturing and water conservancy.

In Section 3, based on GPT-4o and manual verification methods, we analyze the quality of reports generated before and after fine-tuning of QWen and Llama2 models from five language logic and professional perspectives. The analysis process not only analyzes the experimental results from the perspective of the model itself, but also from the perspective of prompt engineering. Combined with all the experimental results in the appendix, it is shown that:

(1) High-quality data is very important to improve the report generation ability of the model in the professional field. Both the fine-tuned QWen and Llama2 model show performance improvement. However, because Qwen model has not only language ability but also arithmetic ability, and covers a wider range of fields and tasks, the performance improvement effect is not obvious after the fine-tuning of the model with a small amount of data. In contrast, Llama2 model is an improved version based on Llama, so the fine-tuning of a small sample of specialized domain data has a great impact on the model, resulting in a more significant performance improvement of Llama2 model after fine-tuning.
(2) Based on the base model of QWen and Llama2, cases in four fields of petroleum and petrochemical, nuclear and thermal power, ocean engineering and energy engineering are asked in both English and Chinese. It is found that Qwen model can better answer Chinese and English questions, which is related to QWen's good generalization ability. Llama2 model can respond better to questions in English. Even when questions are asked in Chinese (Q2 in Table 7), Llama2 model still replies in English. This is because the base model of Llama2 is in English.
(3) QWen's base model will return a short case with traces of a fixed format. But Llama2′s base model requires a human to provide some information to return usable text. These two phenomena prove that Qwen model has stronger generalization ability, and Llama2 model is more suitable for training in professional fields.

Based on the analysis of the generated text from the models, several limitations in the models can be identified:

(1) Language Sensitivity: The Llama2 model is more sensitive to the language in which the question is asked. It produces better results when questions are asked in English but performs poorly, especially with Chinese questioning. This suggests that Llama2 has a strong dependency on the language and is constrained by the base language used in training.
(2) Detail and Accuracy of Generated Content: Both fine-tuned models (QWen and Llama2) improve upon their respective base models, but the accuracy and richness of the content vary depending on the question format. For example, the fine-tuned QWen model produces more accurate and consistent results when asked in the "Question 2″ format, whereas the fine-tuned Llama2 provides more accurate results in the "Question 4" format. The Llama2 model often generates richer and more detailed text, but sometimes the content deviates from the original case, especially when more detailed information is required.
(3) Content Redundancy and Lack of Logic: When the fine-tuned Llama2 model answers in English, the content tends to be redundant, lacking clear logical structure. This is evident when answering "Question 2", where the generated text is numerous but not well-organized.
(4) Model Constraints with Formal and Informal Requirements: The fine-tuned Llama2 model faces challenges in providing content that matches the requirements for both formal and informal tasks. For example, in some cases, the generated text from Llama2 is in English even when the question is posed in Chinese, further highlighting the model's language-specific limitations.
(5) Inconsistencies in Failure Descriptions: While both models provide some correct failure descriptions, the specifics of the failure types (such as cracking, corrosion, and brittle fracture) and the related features (e.g., temperature, pressure) vary between the models. This suggests that the models still struggle with generating consistent failure narratives across different contexts, especially when asked for detailed data or when tasked with creating text that aligns with real-world scenarios.

In summary, while both fine-tuned models show improvements over their base versions, they still suffer from limitations related to language dependency, inconsistency in failure descriptions, redundant content generation, and difficulties in maintaining a structured logical flow. These challenges suggest that further refinement in both language processing and data alignment with real-world cases is required.

5. Conclusion

This paper makes a preliminary exploration in the field of materials based on LLM technology and verifies the effectiveness of establishing databases in nine fields: petrochemical industry, nuclear power and thermal power, ocean engineering, energy engineering, infrastructure, aerospace, transportation, equipment manufacturing, and water conservancy engineering. The ability of LLM to generate failure analysis reports is highly valuable in industrial applications. Using LLMs to achieve the task of generating failure reports can greatly reduce manpower and improve work efficiency. What needs to be studied in the future is as follows:

(1) This study demonstrates that the application of LLMs in various industrial fields is of significant importance, particularly in generating engineering failure analysis reports. Its use in fields like petrochemical, nuclear and thermal power, ocean engineering, and energy engineering can greatly reduce labor costs and improve work efficiency. By automating report generation, LLMs not only accelerate data analysis and enhance accuracy but also provide decision support to engineering sectors, reducing the risks of human error. Thus, this technology has great potential for application in many

industrial domains, especially in areas such as fault detection, quality control, and equipment maintenance.

(2) The approach in this study has some limitations. Firstly, the dataset size is relatively small. Although high-quality manually constructed data is used, the sample size might not be sufficient for the model to perform consistently in more complex or dynamic real-world scenarios. Additionally, due to the failure report generation task being recognized by the machine as a destructive task, the model is highly sensitive to input prompts, and different prompts may lead to significant fluctuations in the output results. To address these issues, future work could involve expanding the dataset, particularly with more diverse samples, to improve the model's generalization ability. Additionally, refining prompt engineering techniques to design more robust input prompts could further enhance model performance.

(3) Through different prompt words, it is found that the report generation ability of LLMs is seriously affected by input prompts. In the future, the application of other LLM models can be further explored to evaluate the performance and advantages or disadvantages of different models in various domain-specific tasks. Secondly, with the expansion of datasets, especially in specialized domains, further validation of LLMs' performance in generating failure analysis reports can be conducted. Finally, prompt engineering techniques can also be a key area for future research, improving the recognition of complex instructions and optimizing input prompts to enhance the model's generation capabilities and accuracy.

Availability of Data and Materials

The dataset for the experiments was uploaded to the author's github repository: https://github.com/jessi-chang/MaterialLLM-result-1.

CRediT authorship contribution statement

Sijie Chang: Writing – original draft, Visualization, Methodology, Data curation, Conceptualization. **Meng Wan:** Validation, Investigation. **Jiaxiang Wang:** Writing – review & editing, Validation. **Hao Du:** Validation, Data curation. **Pufen Zhang:** Validation, Investigation. **Peng Shi:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.rineng.2025.104532.

References

[1] A.M. Olajumoke, I.A. Oke, A.B. Fajobi, M.O. Ogedengbe, Engineering failure analysis of a failed building in Osun State, Nigeria. Journal of failure analysis and prevention 9 (2009) 8–15.
[2] S.M. Zamani, S.A. Hassanzadeh-Tabrizi, H. Sharifi, Failure analysis of drill pipe: a review, Eng. Fail. Anal. 59 (2016) 605–623.
[3] Q. Hu, S. Yang, W. Zhang, G. Da, X. Xu, X. Wang, Corrosion failure analysis of engineering structural steels in tropical marine atmospheres: a comparative study of ordinary and new weathering steels, Eng. Fail. Anal. 156 (2024) 107830.
[4] B. Xia, C. Zhang, Z. Liu, X. Feng, Dynamic crack propagation of ceramic materials under high temperature thermal shock, The International Conference on Computational & Experimental Engineering and Sciences 29 (3) (2024) 1–2, https://doi.org/10.32604/icces.2024.012764.
[5] Y. Pang, P. Li, D. Li, X. Zhou, H. Fan, Q. Wang, Phase-field modeling of thermal shock fracture in functionally graded materials, Eng. Fract. Mech. 307 (2024) 110286.
[6] J.S. Chou, C.K. Chiu, I.K. Huang, K.N. Chi, Failure analysis of wind turbine blade under critical wind loads, Eng. Fail. Anal. 27 (2013) 99–118.
[7] E. Ma, J. Lai, S. Xu, X. Shi, J. Zhang, Y. Zhong, Failure analysis and treatments of a loess tunnel being constructed in ground fissure area, Eng. Fail. Anal. 134 (2022) 106034.
[8] C. Zhang, Y. Zhao, P. Han, Q. Bai, Coal pillar failure analysis and instability evaluation methods: a short review and prospect, Eng. Fail. Anal. 138 (2022) 106344.
[9] P.H. Thike, Z. Xu, Y. Cheng, Y. Jin, P. Shi, Materials failure analysis utilizing rule-case based hybrid reasoning method, Eng. Fail. Anal. 95 (2019) 300–311.
[10] M. Sana, M.A. Ali, S. Ehsan, M. Tlija, A.M. Khan, Investigation of EDM erosion behavior for Ni-based superalloy using experimental and machine learning approach, Mater. Today Commun. (2024) 110819.
[11] K. Jang, G.A. Yun, A self-learning data-driven development of failure criteria of unknown anisotropic ductile materials with deep learning neural network, Computers, Materials and Continua 66 (2) (2021) 1091–1120.
[12] M. Golmohammadi, M. Aryanpour, Analysis and evaluation of machine learning applications in materials design and discovery, Mater. Today Commun. 35 (2023) 105494.
[13] H. Liu, L. Li, On languaging a simulation engine: rapid modeling of nanoporous media sorption by hierarchical language model, Mater. Today Commun. 40 (2024) 109809.
[14] S. Nasiri, M.R. Khosravani, K. Weinberg, Fracture mechanics and mechanical fault detection by artificial intelligence methods: a review, Eng. Fail. Anal. 81 (2017) 270–293.
[15] J. Ding, R. Alroobaea, A.M. Baqasah, A. Althobaiti, R. Miglani, Big data intelligent collection and network failure analysis based on artificial intelligence, Informatica 46 (3) (2022).
[16] A. Manan, K. Kamal, T.A.H. Ratlamwala, M.F. Sheikh, A.G. Abro, T. Zafar, Failure classification in natural gas pipe-lines using artificial intelligence: a case study, Energy Reports 7 (2021) 7640–7647.
[17] W. Schneller, M. Leitner, B. Maier, F. Grün, O. Jantschner, S. Leuders, T. Pfeifer, Artificial intelligence assisted fatigue failure prediction, Int. J. Fatigue 155 (2022) 106580.
[18] Z. Zhang, I. Fidan, M. Allen, Detection of material extrusion in-process failures via deep learning, Inventions 5 (3) (2020) 25.
[19] P. Gavallas, G. Stefanou, D. Savvas, C. Mattrand, J.M. Bourinet, CNN-based prediction of microstructure-derived random property fields of composite materials, Comput. Methods Appl. Mech. Eng. 430 (2024) 117207.
[20] X. Sun, T. Zhou, K. Song, X. Chen, An image recognition based multiaxial low-cycle fatigue life prediction method with CNN model, Int. J. Fatigue 167 (2023) 107324.
[21] Dunn, A., Dagdelen, J., Walker, N., Lee, S., Rosen, A.S., Ceder, G., … & Jain, A. (2022). Structured information extraction from complex scientific text with fine-tuned large language models. arXiv preprint arXiv:2212.05238.
[22] O. Sierepeklis, J.M. Cole, A thermoelectric materials database auto-generated from the scientific literature using ChemDataExtractor, Sci. Data 9 (1) (2022) 648.
[23] M.P. Polak, D. Morgan, Extracting accurate materials data from research papers with conversational language models and prompt engineering, Nat. Commun. 15 (1) (2024) 1569.
[24] Touvron, H., Lavril, T., Izacard, G., Martinet, X., Lachaux, M.A., Lacroix, T., … & Lample, G. (2023). Llama: open and efficient foundation language models. arXiv preprint arXiv:2302.13971.
[25] Touvron, H., Martin, L., Stone, K., Albert, P., Almahairi, A., Babaei, Y., … & Scialom, T. (2023). Llama 2: open foundation and fine-tuned chat models. arXiv preprint arXiv:2307.09288.
[26] Bai, J., Bai, S., Chu, Y., Cui, Z., Dang, K., Deng, X., … & Zhu, T. (2023). Qwen technical report. arXiv preprint arXiv:2309.16609.
[27] K. Mishev, A. Gjorgjevikj, I. Vodenska, L.T. Chitkushev, D. Trajanov, Evaluation of sentiment analysis in finance: from lexicons to transformers, IEEe Access. 8 (2020) 131662–131682.
[28] A. Alonso-Robisco, J.M. Carbó, Analysis of CBDC narrative by central banks using large language models, Financ. Res. Lett. 58 (2023) 104643.
[29] M. Leippold, Thus spoke GPT-3: interviewing a large-language model on climate finance, Financ. Res. Lett. 53 (2023) 103617.
[30] V.R. Kukkala, S.P. Praveen, N.S.K.M.K. Tirumanadham, P.N. Srinivasu, A study on outlier detection and feature engineering strategies in machine learning for heart disease prediction, Computer Systems Science & Engineering 48 (5) (2024).
[31] C. Peng, X. Yang, A. Chen, K.E. Smith, N. PourNejatian, A.B. Costa, Y. Wu, A study of generative large language model for medical research and healthcare, NPJ. Digit. Med. 6 (1) (2023) 210.
[32] L. Peng, G. Luo, S. Zhou, J. Chen, Z. Xu, J. Sun, R. Zhang, An in-depth evaluation of federated learning on biomedical natural language processing for information extraction, NPJ. Digit. Med. 7 (1) (2024) 127.
[33] S.K. Srivastava, S. Routray, S. Bag, S. Gupta, J.Z. Zhang, Exploring the potential of large language models in supply chain management: a study using big data, Journal of Global Information Management (JGIM) 32 (1) (2024) 1–29.

[34] E. Kasneci, K. Seßler, S. Küchemann, M. Bannert, D. Dementieva, F. Fischer, G. Kasneci, ChatGPT for good? On opportunities and challenges of large language models for education, Learn. Individ. Differ. 103 (2023) 102274.
[35] K. Mahowald, A.A. Ivanova, I.A. Blank, N. Kanwisher, J.B. Tenenbaum, E. Fedorenko, Dissociating language and thought in large language models, Trends Cogn. Sci. (Regul. Ed.) (2024).
[36] Zhao, W.X., Zhou, K., Li, J., Tang, T., Wang, X., Hou, Y., … & Wen, J.R. (2023). A survey of large language models. arXiv preprint arXiv:2303.18223.
[37] Y. Hu, Y. Goktas, D.D. Yellamati, C. De Tassigny, The use and misuse of pre-trained generative large language models in reliability engineering, in: 2024 Annual Reliability and Maintainability Symposium (RAMS), IEEE, 2024, pp. 1–7.
[38] Stewart, M., Hodkiewicz, M., & Li, S. (2023). Large language models for failure mode classification: an investigation. arXiv preprint arXiv:2309.08181.
[39] G. Jiang, Z. Ma, L. Zhang, J. Chen, Prompt engineering to inform large language model in automated building energy modeling, Energy (2025) 134548.
[40] Y. Qi, X. Zhao, S. Khastgir, X. Huang, safety analysis in the era of large language models: a case study of STPA using ChatGPT, Mach. Learn. Appl. (2025) 100622.
[41] Pu, G., Jain, A., Yin, J., & Kaplan, R. (2023). Empirical analysis of the strengths and weaknesses of PEFT techniques for LLMs. arXiv preprint arXiv:2304.14999.
[42] P.M. de Castro, A.A. Fernandes, Methodologies for failure analysis: a critical survey, Mater. Des. 25 (2) (2004) 117–123.